

Towards a Unified Framework for Responsible Social Computing Undergraduate and Graduate Education

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Abstract

Social computing education is gaining traction across computer science and information science disciplines, yet lacks a standardized pedagogical framework for topics to teach, as well as strategies for engaging students. This workshop aims to convene experts in CSCW to develop a cohesive framework for undergraduate and graduate social computing courses. Emphasizing ethical and cultural considerations, the workshop will explore core topics spanning sociological, psychological, technical, and design aspects of social computing. Through interactive sessions, attendees will collaboratively shape a curricular framework that is responsive to diverse country contexts and ethical imperatives. At the conclusion of the workshop, we anticipate we will publish a framework for topics and pedagogical approaches to social computing, as well as a repository of existing courses.

CCS Concepts

- Human-centered computing~Collaborative and social computing~ Collaborative and social computing theory, concepts, and paradigms

Keywords

social computing, undergraduate and graduate education, responsible computing education

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1 Introduction and Workshop Goals

Social computing is a core topic in human-centered computing [1], and there has been a growing interest in offering interdisciplinary Social Computing courses at the graduate or undergraduate level in the field of Computer Science and Information Science. However, the scope of social computing courses, key topics, and pedagogical approaches to teaching vary widely from course to course. For example, there is no unifying textbook that details core knowledge and competencies students should have as they learn about social computing. Further, courses can take on different angles to cover relevant content – such as psychological and sociological foundations, technical implementation, design, and community-centered technologies. The goal of this workshop is to bring together the CSCW community to discuss the core knowledge and competencies related to social computing, and consider the nuances of how these courses might need to be flexible to different contexts, while emphasizing responsible and culturally-sensitive design [2]. In particular, much of the discourse surrounding social computing is focused on North American and European contexts, and doesn't take into account the perspectives, identities, and values of instructors and learners from other country contexts [3].

Towards that end, the goal of this workshop is to elucidate a framework for social computing undergraduate & graduate courses, where country-context & ethical implications are core to all parts of the framework. Our framework will revolve around core topics in social computing, centering the sociological and psychological, technical, and design aspects of the field, and how these topics might shift according to country context. Our framework will discuss how ethics and responsible design is woven into all parts of the curriculum, rather than being a standalone topic. Finally, we will accompany the framework with pedagogical approaches for teaching said topics, such as project-based learning and community-engaged coursework. In addition to the framework, we aim to create a community-run repository of social computing courses, such that others who wish to teach such courses in the future can have access to tested resources.

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Thus, at the culmination of the workshop, we will produce: 1) a framework for core topics and competencies in social computing that span sociological and psychological, technical, and design aspects, 2) definition of how the framework is responsive to ethical and country-context considerations, 3) pedagogical resources for teaching said topics, and 4) a repository of related courses, with mechanisms for people to add to the list. We plan to document our process for creating the framework, with the future aim of a publication on this topic.

2 Schedule and Activities

Our workshop will consist of expert presentations and interactive brainstorming and meaning making activities in order to achieve the proposed goals.

Morning Session (10 AM - 12 PM)

- (20 minutes) Introduction - we will introduce workshop attendees to the topic of social computing and introduce our initial framework of social & psychological, technical, and design elements of social computing, with ethics and country considerations interwoven
- (30 minutes) Attendee introductions - We will use a social identities wheel to discuss who workshop attendees are as it relates to core social computing topics, and how they approach the field of social computing. For example, ethnicity and country origin are core to our social identities. A person might introduce the unique ways they have seen social computing affect healthcare in their country. Alternatively, someone might include their profession and field as their social identity (e.g., psychologist), which informs the theoretical frameworks with which they approach social computing.
- (50 minutes) Individual presentations - Approximately five presenters will speak on their social computing course, learnings from how to teach the course, and ideas for a unified framework. During these presentations, workshop attendees will add notes to a shared whiteboard on concepts or ideas that they wish to discuss further in the afternoon session.
- (20 minutes) unallocated time for breaks

Afternoon Session (1 PM - 3 PM)

- (50 minutes) Small group affinity diagramming - The attendees will divide into groups of three to four, and affinity diagram [4] the notes from the morning session. The goal is for each group to discover relationships between the themes discussed in the morning session, and propose organization of the framework.
- (50 minutes) Share out - Each team will engage in a structured share out of their affinity diagram. The attendees will engage in a guided discussion of overarching themes, and resources, such that we create our final framework.

- (20 minutes) unallocated time for breaks

3 About the Organizers

Dr. Angela E.B. Stewart is an Assistant Professor in the University of Pittsburgh School of Computing and Information and Research Scientist at the Learning Research and Development Center. Prior to that, she was a Postdoctoral Fellow in the Human-Computer Interaction Institute at Carnegie Mellon University. She received a PhD in Computer Science from the University of Colorado Boulder (2020) and a Bachelor of Software Engineering from Auburn University (2015). Angela conducts research at the intersection of the learning sciences, artificial intelligence, and human-computer interaction. She uses multimodal data to understand students' social and cognitive states, particularly in collaborative STEM learning. She also creates equitable educational spaces by designing technologies that support the agency of students and teachers. Angela applies a culturally-responsive lens to her research, with a particular focus in emboldening Black girls' design of transformative technologies. Angela was named a 2021 - 2022 Emerging Scholar by the International Society of the Learning Sciences, and recently received an NSF Racial Equity in STEM Award for creating an intersectional AI learning ecosystem for Black girls. Angela has taught an undergraduate social computing course since joining Pitt, with a focus on using frameworks of power and identity to understand social technologies.

Dr. Rosta Farzan is the Associate Dean for Diversity, Equity, and Inclusion in the School of Computing and Information, and an Associate Professor in the Department of Informatics and Networked Systems. She completed her PhD in Intelligent Systems Program at the University of Pittsburgh in 2009. She joined the faculty of School of Computing and Information in 2012. She has been teaching a Social Computing course for over ten years as well as doing research in the field. Her research focuses on partnering with local marginalized communities to design, implement, and deploy technology that centers individuals and communities that have been historically excluded from the space. Her work has been funded by federal agencies, foundations, and industry and has resulted in publications in high impact venues in the field of Human-Computer Interaction, Social Computing, and Information Science, including a number of best paper awards at major conferences. Her contribution to community-engaged scholarship was awarded by the University in 2022 Chancellor Distinguished Service award.

Dr. Catherine Delcourt is an Assistant Professor in Computer Science at Wellesley College. Her research focuses on methods for prototyping large scale social systems and participatory methods for reimagining more positive online spaces with youth. At Wellesley, she teaches courses across the CS curriculum and created an upper-level Social Computing course that introduces students to core social theories central to Social Computing research, and students collaborate in a semester-long group

project designing, building, and evaluating a functional social system.

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